Network modeling - AY 2026/2027
Homework 1 - Source model

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Abstract

I characterize outbound UDP traffic generated by Discord Desktop during a voice-channel session. Packet inter-arrivals are nearly periodic while active, with a small number of long gaps. I therefore model the source as an On–Off packet process, use an empirical distribution for frame lengths, and compare a fixed-seed synthetic trace with the observed trace.

1 Measurement

I captured traffic with Wireshark on Windows 10 while Discord Desktop was idle, sending text messages, in a voice channel, and with camera or screen sharing enabled. I recorded approximate phase times. I analyze only the nominal speaking interval (160–210 s) and one direction: local host to remote server, UDP stream 94 (local port 62392 to 104.29.146.41:19305). After filtering by stream and direction, I retained 1,357 packets from 179.031 to 209.996 s, i.e., 30.965 s of observed traffic. The stream is not present over the full nominal 50 s speaking interval, so my results refer only to this selected segment.

For each packet, I define its capture time as an event time τ_i, $A_i = \tau_i - \tau_{i-1}$, $B_i = 1$, and $W_{i,1} = L_i$, where L_i is `frame.len` in bytes. This packet-event definition avoids inferring application-layer batches from encrypted traffic. Frame length includes captured network headers and is not the encoded audio payload. I compute the observed mean uplink rate as $8 \sum_i L_i / T$.

2 Source Model

The 1,356 packet gaps have mean 22.836 ms, median 19.995 ms, and coefficient of variation (CV) 1.863. A single exponential model is not appropriate: most gaps are tightly concentrated around 20 ms, but a few long gaps inflate the mean and variance. I classify gaps above three times the median (59.986 ms) as Off periods. The remaining 1,343 On gaps have mean 19.854 ms and CV 0.102. Their moment-matched Gamma model has shape $k = 96.80$ and scale $\theta = 0.205$ ms. I resample the 13 Off gaps empirically because this sample is too small to justify a parametric tail fit. A threshold sensitivity check at 40, 60, and 80 ms yields 14, 13, and 13 Off gaps, respectively; On-gap CV remains between 0.094 and 0.102.

Observed frame lengths have mean 88.85 B, median 89 B, standard deviation 3.04 B, and range 50–116 B. I model $W_{i,1}$ by empirical resampling and keep $B_i \equiv 1$. I resample synthetic On-run lengths and Off gaps from the observed sequence and draw On inter-arrivals from the fitted Gamma law. A fixed random seed ensures reproducibility. I sample packet sizes independently, so the model does not preserve dependence between packet size, inter-arrival, and activity state.

3 Results

I set the synthetic observation window to the same 30.965 s as the measured segment. It contains 1,384 packets and carries 31.723 kbit/s, compared with 31.150 kbit/s observed (1.84% higher). The packet count is a stochastic outcome and is not constrained to equal the empirical count. Figure 1 compares temporal scatterplots and CDFs of inter-arrival times and packet lengths, together with bit-rate traces on 20 ms, 100 ms, and 1 s windows. The regular active spacing and observed frame-size concentration are visible in both traces; the long-gap tail remains sensitive to the small empirical Off sample.

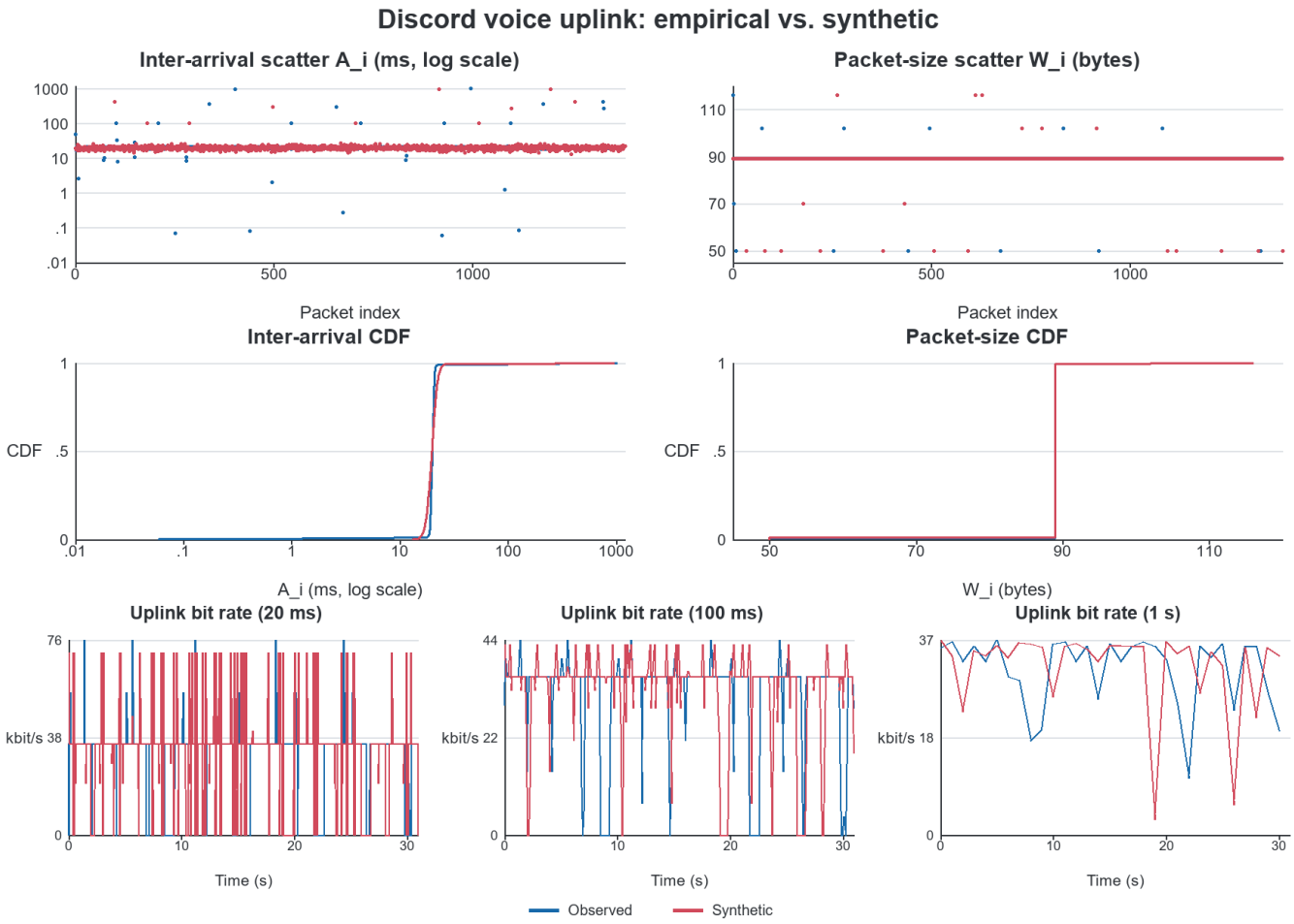


Figure 1: Observed and synthetic voice-uplink traces: temporal scatterplots, CDFs of A_i and W_i, and uplink bit rate on 20 ms, 100 ms, and 1 s windows.

4 Discussion

My results support near-periodic packet generation during active speech plus occasional long gaps; a single exponential inter-arrival model would hide this structure. I use the 60 ms state threshold as a transparent heuristic; its classification is stable from 60 to 80 ms but changes slightly at 40 ms. Because my capture interval is short and my phase boundaries are approximate, long gaps may reflect silence suppression, capture or network effects, or phase timing. The encrypted capture cannot identify codec frames or establish that each UDP datagram is one audio frame. I therefore interpret the fitted source as a model of packets observed at the host interface, not of Discord's internal audio encoder.

As an exploratory comparison, I examined UDP stream 150, a possible screen-sharing flow. Within the approximate 295–355 s screen-sharing phase, it carries 356 packets from 315.12 to 350.07 s (34.95 s between first and last packet): 60.01 kbit/s over that active span, or 34.96 kbit/s averaged over the full 60 s phase. Packet lengths average 736.5 B (median 1,121 B; range 85–1,292 B), and packet inter-arrivals have CV 6.79. Grouping packets separated by at most 3 ms gives 96 bursts with mean size 3.71 packets; thresholds of 1 and 5 ms give 121 and 89 bursts. The remaining 35 packets of this flow occur after the nominal phase, from about 365 to 375 s, so I exclude them from these phase-specific statistics. Because phase times are approximate and payloads are encrypted, I treat the flow as a plausible but unconfirmed screen-sharing candidate and do not fit a second source model.